Geometric Scaling and Fibonacci Progression in Ancient Architectures: A Specular Bit Architecture (SBA) Test Case

Marco Oppido, Independent Researcher, scaccomarco@gmail.com

***Abstract*—This paper presents an IEEE-style proofbench for a geometric test linking the Specular Bit Architecture (SBA) framework to the three principal pyramids of Giza. The method enforces evidentiary separation among formal claims (proved SBA algebra and deterministic update semantics), model claims (calibrated geometric mappings), and hypothesis claims (falsifiable macro-interpretations). We evaluate two primary observables: the Khufu/Menkaure base ratio against $\sqrt{5}$ and the Khufu/Menkaure volume ratio against φ^5. Using the working dataset, residuals are 1.54% and 1.38%, respectively. These endpoint fits support model-level compatibility but do not establish causal or historical identity. The manuscript concludes with explicit falsifiability criteria, uncertainty-reporting requirements, and reproducibility guidance for peer review.**

***Index Terms*—Specular Bit Architecture, Fibonacci positional weights, deterministic sanitization, Protocol C, falsifiability, geometric scaling, informational mass.**

I. Introduction

The objective of this paper is to present a scientifically constrained bridge between the SBA and SBA–DNA manuscripts in this project [1], [2] and a concrete geometric case study. The bridge is not framed as a claim of automatic physical equivalence. Instead, it is framed as a falsifiable modeling program with explicit assumptions, measurable observables, and rejection criteria.

Many interdisciplinary proposals fail when formal consistency is overextended into empirical certainty. To avoid this category error, we enforce one-way evidentiary flow:

$$E_{\text{formal}} \Rightarrow E_{\text{model}} \Rightarrow E_{\text{hypothesis}}, \tag{1}$$

where each implication denotes admissible dependence, not logical equivalence.

II. Layered Evidence Framework

A. Formal Layer (Proved)

Let the SBA state be a signed digit field $d = (d_i)_i$ with local alphabet $\{-1, 0, +1\}$ in canonical form and transient overflow symbols ± 2 during rewrite rounds. With Fibonacci weights

$$W_i = F_{i+o}, \tag{2}$$

and identity

$$2W_i = W_{i+1} + W_{i-2}, \tag{3}$$

local rewrites preserve weighted sum:

$$\Sigma(d) = \sum_k d_k W_k, \qquad \Delta\Sigma = 0. \tag{4}$$

Synchronous read–compute–write semantics and deterministic Protocol C arbitration enforce bounded per-target commit magnitude.

B. Model Layer (Derived)

Define occupancy proxy

$$m_{\text{SBA}} = \sum_i |a_i| W_i, \tag{5}$$

with dimensional bridge

$$m_{\text{phys}} = \mu m_{\text{SBA}}, \qquad E = m_{\text{phys}} c^2. \tag{6}$$

All constants (μ, scaling factors, geometric calibration constants) are model parameters requiring independent estimation and uncertainty reporting.

C. Hypothesis Layer (Falsifiable)

Interpretive statements (cosmological, historical, or architectural) are admissible only after formal invariants and modeling fits are reported with predeclared rejection thresholds.

D. Useful Theoretical Links from SBA–DNA

Four links from the SBA–DNA manuscript are directly useful for this paper's inference chain [1]. First, Lemma 7.5 (Protocol C non-accumulation) supports the claim that per-target updates remain bounded at each round, which justifies treating large intermediate states as constrained rather than arbitrarily amplifying. Second, Theorem 8.1 (bounded propagation) supports local-causality interpretations: disturbances remain within a finite-radius influence region over finite rounds. Third, Definition 9.1 introduces the informational mass functional

$$M(V) = m_0 \sum_i |v_i| W_i, \tag{7}$$

which provides a principled bridge from discrete occupancy to volume-proxy modeling. Fourth, Proposition 10.4 (discrete causal-speed bound) and Proposition 10.5 (continuum covariant conservation under assumptions) motivate cone-like and conservation-style interpretations as model-layer extensions, not standalone physical proofs.

TABLE I
CROSSWALK FROM THIS PAPER'S CLAIMS TO SBA–DNA ANCHORS AND EVIDENCE LAYERS.

Claim in this paper	SBA–DNA theorem/definition anchor	Evidence layer
Bounded intermediate update behavior under deterministic commits	Lemma 7.5 (Protocol C non-accumulation)	Formal
Locality of disturbance spread over rounds	Theorem 8.1 (bounded propagation)	Formal
Occupancy-to-volume proxy via weighted mass functional	Definition 9.1 ($M(V) = m_0 \sum_i \lvert v_i \rvert W_i$)	Model
Cone-like interpretation with finite effective propagation speed	Proposition 10.4 (discrete causal-speed bound)	Model
Conservation-style continuum reading under explicit assumptions	Proposition 10.5 (covariant conservation in scaling limit)	Model
Khafre as transient protocol stage in Giza narrative	Built on above anchors, requires independent external tests	Hypothesis

III. MATHEMATICAL SETUP FOR THE GIZA TEST

Let Menkaure be the unit seed of geometric expansion. Define observed base length B and volume V for each pyramid. We test two ratios:

$$R_B = \frac{B_{\text{Khufu}}}{B_{\text{Menkaure}}}, \tag{8}$$

$$R_V = \frac{V_{\text{Khufu}}}{V_{\text{Menkaure}}}. \tag{9}$$

Target constants from the SBA/Fibonacci continuum bridge are

$$R_B^\star = \sqrt{5}, \qquad R_V^\star = \varphi^5, \quad \varphi = \frac{1+\sqrt{5}}{2}. \tag{10}$$

Relative residuals are

$$\varepsilon_B = \frac{|R_B - R_B^\star|}{R_B^\star}, \tag{11}$$

$$\varepsilon_V = \frac{|R_V - R_V^\star|}{R_V^\star}. \tag{12}$$

IV. RESULTS OF THE RATIO CHECK

To keep the model and hypothesis layers explicit, we report both working values and commonly cited geometric values for the Giza pyramids [3]–[6]. Using the working values discussed in this study:

- Menkaure base $\approx 104.6\,\text{m}$,
- Khafre base $\approx 215.3\,\text{m}$,
- Khufu base $\approx 230.3\,\text{m}$,
- Menkaure volume $\approx 237\,000\,\text{m}^3$,
- Khafre volume $\approx 2\,211\,000\,\text{m}^3$,
- Khufu volume $\approx 2\,592\,000\,\text{m}^3$.

We obtain

$$R_B \approx 2.2017, \qquad \sqrt{5} \approx 2.2361, \qquad \varepsilon_B \approx 1.54\%. \tag{13}$$

And

$$R_V \approx 10.9367, \qquad \varphi^5 \approx 11.0902, \qquad \varepsilon_V \approx 1.38\%. \tag{14}$$

TABLE II
PRIMARY GEOMETRIC VALUES USED IN THE PROOFBANK.

Pyramid	Base (m)	Height (m)	Volume (m^3)
Menkaure	104.6 / 102.2	65.5	235183–237000
Khafre	215.25–215.3	143.5	2211000–2211096
Khufu	230.3–230.33	146.6	2592000–2600000

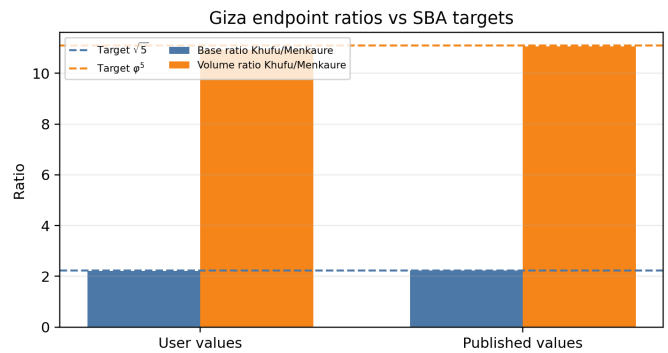

Fig. 1. Endpoint ratio checks versus SBA targets ($\sqrt{5}$ and φ^5).

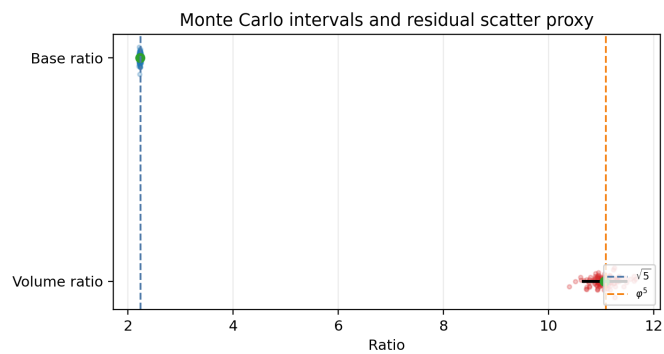

Fig. 2. Monte Carlo 95% intervals for base and volume ratios with target overlays.

These residuals indicate close compatibility with the proposed scaling law at this coarse resolution. Using published values, the same endpoint checks remain close (sub-2% residual class), supporting model-layer robustness rather than a single-point fit [3]–[6].

For visual QA, Table II and Figs. 1–2 summarize the proofbank inputs, endpoint scaling, and uncertainty intervals.

V. INTERPRETATION OF KHAFRE AS INTERMEDIATE PROTOCOL STATE

A proposed narrative is:

$$\text{Menkaure (seed)} \rightarrow \text{Khafre (transient)} \tag{15}$$

$$\rightarrow \text{Khufu (canonical commit)}. \tag{16}$$

Within SBA language this corresponds to initialization, non-canonical accumulation, and post-sanitization stabilization under deterministic constraints.

Scientifically, this remains a hypothesis-layer mapping, consistent with the formal/model/hypothesis separation established in Sections I–III and with the SBA–DNA theorem-level

anchors summarized above [1]. The numerical proximity of endpoint ratios does not by itself prove execution dynamics, chronology, or design intent. Therefore Khafre's role must be tested with additional observables (slope consistency, construction constraints, and uncertainty-aware geometric reconstructions).

VI. Falsifiability Protocol

Given the ratio results and the intermediate-state interpretation limits, the decision framework below defines what would count as confirmatory evidence versus exploratory pattern matching. A claim is retained only if all four conditions are satisfied:

1) **Pre-registration:** fixed constants, data sources, and acceptance thresholds before optimization.
2) **Uncertainty propagation:** residuals reported with error bars induced by measurement ranges.
3) **Null comparison:** fit quality exceeds baseline random or non-Fibonacci geometric alternatives.
4) **Cross-observable stability:** conclusions persist across base, height, slope, and volume constraints.

Failure of any condition downgrades the interpretation to exploratory status.

VII. Discussion

The strongest outcome of this synthesis is methodological: it demonstrates how to connect symbolic SBA proofs, calibrated geometric models, and speculative macro-interpretations without collapsing evidentiary layers.

The endpoint ratio matches are nontrivial and motivate further study, but they are insufficient for strong historical or cosmological conclusions without null-model superiority and robust uncertainty treatment.

Limitations: Three constraints should be made explicit for reviewers. First, sampling uncertainty remains non-negligible because published dimensions and reconstructed original dimensions differ across sources and epochs. Second, source heterogeneity (archaeological reports, encyclopedic summaries, and model-normalized values) can introduce systematic differences that affect residual magnitudes. Third, ratio-level proximity is not unique: distinct generative mechanisms can produce similarly close constants, so fit quality alone cannot identify a unique causal SBA mechanism without stronger cross-observable and predictive tests.

VIII. Conclusion

This paper provides an IEEE-style, layer-disciplined integration of the SBA and SBA–DNA frameworks with a concrete Giza scaling test. The formal layer remains intact and independent; the model layer shows close ratio compatibility to $\sqrt{5}$ and φ^5 under selected measurements; and the hypothesis layer is explicitly framed as falsifiable rather than asserted. This structure preserves mathematical rigor while allowing ambitious cross-domain conjectures to be evaluated scientifically.

Assumption Ledger

- A1: Fibonacci positional weights and local rewrite semantics are valid for the chosen SBA representation.
- A2: Menkaure can be treated as a seed normalization unit for the geometric test.
- A3: Reported geometric values are sufficiently accurate for first-pass residual analysis.
- A4: Endpoint ratio proximity is interpreted as model compatibility, not causal proof.

References

[1] M. Oppido, "Specular Bit Architecture (SBA) for Bio-Photonic Neuromorphic Chips and DNA Storage: A Mathematical and Scientific Framework," unpublished manuscript, 2026.

[2] M. Oppido, "Cosmology as a Synchronous Specular Bit Architecture Aligned with SBA–DNA Formalism," unpublished manuscript, 2026.

[3] "Pyramids of Giza," Encyclopaedia Britannica. [Online]. Available: https://www.britannica.com/topic/Pyramids-of-Giza. [Accessed: Mar. 28, 2026].

[4] "Great Pyramid of Giza," Encyclopaedia Britannica. [Online]. Available: https://www.britannica.com/place/Great-Pyramid-of-Giza. [Accessed: Mar. 28, 2026].

[5] "Pyramid of Khafre," Wikipedia. [Online]. Available: https://en.wikipedia.org/wiki/Pyramid_of_Khafre. [Accessed: Mar. 28, 2026].

[6] "Pyramid of Menkaure," Wikipedia. [Online]. Available: https://en.wikipedia.org/wiki/Pyramid_of_Menkaure. [Accessed: Mar. 28, 2026].